

HR ANALYTICS FOR INSURANCE

Agent Performance, Productivity, and Retention Analysis

Field	Details
Program	Data Analytics
Project Title	HR Analytics for Insurance: Agent Performance and Retention Analysis
Project Type	Data Analytics Project (Excel-Based)
Dataset Used	Agent Master Data, Activity Log Data, Performance and Retention Data
Tools Used	Microsoft Excel (Pivot Tables, Charts, Formulas)

A Beginner-Level Data Analytics Project Report

Prepared Using Microsoft Excel

Submitted by:
Roshitha
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1. Introduction to HR Analytics in Insurance

1.1 What is HR Analytics?

HR Analytics (also called People Analytics or Workforce Analytics) is the practice of using data and statistical methods to understand how employees work and how to improve their performance and experience. Instead of making decisions based on guesswork or gut feeling, HR analytics uses actual data to guide management decisions.

In simple terms, HR analytics answers questions like: Which agents are performing well? Why are some agents leaving the company? What training will help agents sell more policies?

By collecting and analysing data on attendance, sales, activity levels, and retention, HR managers and team leaders can take smarter, faster, and more effective actions.

1.2 Importance of HR Analytics in the Insurance Industry

The insurance industry is one of the most agent-dependent industries in the world. Unlike a software company where a product can be sold online, insurance policies require human interaction. An insurance agent builds trust with customers, explains policy benefits, handles objections, and closes sales.

This makes the performance and retention of agents extremely critical for the success of any insurance company. HR analytics plays a very important role in this context for the following reasons:

- **Performance Visibility:** Companies can see exactly how each agent is performing in terms of calls made, meetings held, proposals sent, and policies sold.
- **Early Warning System:** If an agent's activity suddenly drops, analytics can flag this before the agent quits — giving managers time to intervene.
- **Data-Driven Training:** Instead of giving the same training to all agents, analytics helps identify which specific skill (communication, closing, pitching) an agent needs to improve.
- **Revenue Optimisation:** By identifying top performers and understanding what they do differently, companies can replicate success across the team.
- **Cost Reduction:** Replacing an agent is expensive (recruitment, training, lost productivity). Retaining agents through better HR decisions saves money.

1.3 Role of Insurance Agents in Revenue Generation

Insurance agents are the primary revenue generators in any insurance company. They are responsible for:

- Reaching out to potential customers (cold calls, referrals, walk-ins)
- Scheduling meetings and explaining policy benefits
- Converting meetings into proposals and proposals into sales
- Collecting First Year Premiums (FYP) which form the company's direct revenue

In our dataset, the metric used to measure revenue is FYP — First Year Premium. This is the total premium amount paid by a customer when they buy a policy for the first time. An agent's FYP directly reflects how much business they have brought to the company.

The activity funnel — Calls → Meetings → Proposals → Sales → FYP — is the backbone of insurance revenue generation, and HR analytics helps monitor and improve every stage of this funnel.

2. Problem Statement

The insurance industry faces two major HR challenges that directly affect business performance: high attrition among agents and uneven productivity. This project focuses on understanding and addressing both of these challenges using data.

2.1 High Attrition Among Insurance Agents

Agent attrition (or turnover) is one of the biggest problems in the insurance sector. Studies suggest that the insurance industry has one of the highest attrition rates among all industries — with some companies losing 30% to 50% of their agents every year.

In our dataset of 130 agent records, we can already see signs of this problem. Out of approximately 30 unique agents in the Agent Master sheet, many are marked as 'Inactive'. The Retention analysis sheet shows that while most agents fall under 'Low Risk', a group of 16 agents out of 130 records fall under 'Medium Risk', indicating they are on the path to disengagement.

The key reasons for high attrition in insurance include:

- Low earnings in the early months (before building a client base)
- Lack of proper training and support from managers
- Frustration due to repeated rejections during cold calls
- Feeling of isolation — agents often work independently
- Lack of recognition for effort even when sales are low

2.2 Low Productivity and Uneven Performance

The second major problem is that even among active agents, there is a huge gap in productivity. Our Performance sheet data clearly shows this:

- 50% of agents are classified as Low Performers (FYP below median)
- 40% of agents are Average Performers
- Only 10% of agents are Star Performers (top 10% by FYP)

This means that a small group of star agents is responsible for driving a disproportionately large share of the company's total revenue. If even one or two of these top agents leave, the company faces a significant revenue loss.

The Activity Log data also reveals where agents are losing business in the funnel. The most common problems identified are:

- Poor Communication (46 cases out of 130) — agents are not able to convert enough calls into meetings
- Poor Closing (31 cases) — agents have meetings but cannot convert them into sales
- Poor Pitching (17 cases) — agents present proposals but fail to close deals

These specific problem areas show that the issue is not just about effort — it is about skill gaps at different stages of the sales funnel.

3. Objectives of the Project

Based on the problem statement above, this project was designed with three clear objectives:

3.1 Improve Productivity

The first objective is to identify where agents are losing business in the sales funnel and suggest ways to improve conversion rates at each stage. By analysing the activity log data, we aim to:

- Calculate conversion rates at each stage: Calls → Meetings, Meetings → Proposals, Proposals → Sales
- Identify funnel leakage points — where the biggest drop-off happens
- Suggest targeted interventions to plug these leaks

3.2 Identify Performance Gaps

The second objective is to segment agents into performance categories (Star, Average, Low) and understand the characteristics of each group. This helps managers focus their attention and resources more effectively:

- Star Performers: Understand what they are doing right so others can learn from them
- Average Performers: Identify what is holding them back and provide coaching
- Low Performers: Determine whether the issue is skill, motivation, or a structural barrier

3.3 Predict Retention Risk

The third objective is to predict which agents are at risk of leaving the company before they actually leave. By using activity data (meetings last month, sales last month), we classify agents into three retention risk levels:

- Low Risk: Agent is active and likely to stay
- Medium Risk: Agent's activity has dropped and they may disengage
- High Risk: Agent has very low activity and is likely to leave

This early warning system allows managers to intervene with at-risk agents before it is too late — saving both the agent and the company's investment in them.

4. Data Description

The dataset used in this project is contained in a single Microsoft Excel workbook titled 'HR_Analytics_for_Insurance.xlsx'. The workbook contains multiple sheets, each serving a specific purpose. Below is a detailed description of each data source.

4.1 Agent Master Sheet

This is the primary reference sheet containing basic details of all agents. It acts like a database of all agents who have ever joined the company.

Column Name	Data Type	Description	Example Values
AgentID	Text (ID)	Unique identifier for each agent	A101, A102, A130
Name	Text	Full name of the agent	Latha, Arjun, Divya
JoinDate	Date	Date when the agent joined the company	01-Jan-2024
LicenseDate	Date	Date when the agent got their license	10-Jan-2024
Status	Text (Category)	Whether the agent is Active or Inactive	Active, Inactive
Region	Text (Category)	Geographic zone the agent belongs to	North, South, East, West
Source	Text (Category)	How the agent was recruited	Online, Walk-in, Referral

The Agent Master sheet contains 30 unique agents (A101 to A130), with some agents appearing in multiple months' records. This sheet is used for demographic analysis and segmentation by region and source.

4.2 Activity Log Sheet

This is the most important analytical sheet in the workbook. It records the monthly activity of each agent — how many calls they made, how many meetings they held, how many proposals they submitted, and how many policies they sold.

Column Name	Description	Example / Formula
AgentID	Links to Agent Master	A101
Month	Month of activity	Jan, Feb, Mar...
Calls	Number of calls made (random range 70–160)	RANDBETWEEN(70,160)
Meetings	Meetings held with clients (range 5–50)	RANDBETWEEN(5,50)

Column Name	Description	Example / Formula
Proposals	Proposals sent to clients (range 2–25)	RANDBETWEEN(2,25)
Sales	Policies sold (range 1–15)	RANDBETWEEN(1,15)
FYP	First Year Premium collected (₹8,000–₹60,000)	RANDBETWEEN(8000,60000)
Meeting Rate	Meetings ÷ Calls (conversion ratio)	=Meetings/Calls
Proposal Rate	Proposals ÷ Meetings	=Proposals/Meetings
Sales Conversion	Sales ÷ Proposals	=Sales/Proposals
Problem	Auto-identified issue based on rates	Poor Communication / Closing / Pitching / Strong

The Activity Log records 130 entries (5 months × approximately 26 agents per month), providing a rich view of agent performance trends over time.

4.3 Performance Sheet

This sheet calculates the total FYP earned by each agent across all months and classifies them into performance tiers using Excel's PERCENTILE function.

- Total FYP ranges from ₹87,974 (lowest) to ₹214,870 (highest) across the 30 agents
- Median FYP (50th percentile): ₹1,52,424
- Top performer threshold (90th percentile): ₹1,93,090
- Performance classification: Star (top 10%), Average (above median), Low (below median)

4.4 Retention Sheet

This sheet evaluates the retention risk of each agent based on two key indicators from the previous month: number of meetings held and number of sales made. The logic is straightforward and easy to understand:

Retention Risk Logic (Excel IF Formula)

- HIGH RISK: Meetings < 5 AND Sales = 0 (very low activity + no earnings)
- MEDIUM RISK: Meetings < 10 (moderate activity, may be struggling)
- LOW RISK: Meetings ≥ 10 (active and likely engaged)

The Retention sheet contains 130 records. The summary shows: 114 Low Risk records (87.7%) and 16 Medium Risk records (12.3%). Encouragingly, there are no High Risk agents in the current dataset.

4.5 Insight → Action Sheet

This sheet uses Pivot Tables to summarise the 'Problem' column from the Activity Log. It shows how many times each problem type occurred and which agents have recurring issues. The summary counts are:

Problem Category	Count	Percentage of Total	Business Implication
Poor Communication	46	35.4%	Agents struggle to convert calls to meetings
Strong Performer	36	27.7%	No significant problem — performing well
Poor Closing	31	23.8%	Agents meet clients but cannot convert to sales
Poor Pitching	17	13.1%	Proposals sent but deals not closed
Grand Total	130	100%	Complete dataset coverage

5. Data Preparation

Before any analysis can be done, raw data must be cleaned, validated, and enriched with calculated columns. This section explains all the data preparation steps followed in Excel.

5.1 Data Cleaning Steps

Data cleaning is the process of fixing errors, removing inconsistencies, and making the data ready for analysis. The following steps were performed:

Step 1: Checking for Duplicate Records

We used Excel's 'Remove Duplicates' feature (Data tab → Remove Duplicates) to check if any AgentID appeared more than once in the same month. Since the dataset uses RANDBETWEEN formulas, duplicates are inherently avoided, but the check was performed as a best practice.

Step 2: Verifying Status Values

The Status column in the Agent Master sheet should only contain 'Active' or 'Inactive'. A quick filter check confirmed that no other values existed. Out of 30 agents, 16 are Active and 14 are Inactive, giving an attrition indicator of approximately 47%.

Step 3: Checking Date Consistency

The JoinDate and LicenseDate columns were verified to ensure that no agent had a LicenseDate before their JoinDate. In all cases, the LicenseDate was 9 days after the JoinDate, which is logical (time needed to process the license).

Step 4: Validating Numeric Ranges

Activity values (Calls, Meetings, Proposals, Sales, FYP) were checked using MIN and MAX functions to ensure they fell within the expected ranges defined in the RANDBETWEEN formulas.

Column	Expected Range	Validation Method
Calls	70 – 160	=MIN(Calls column) and =MAX(Calls column)
Meetings	5 – 50	=MIN(Meetings) and =MAX(Meetings)
Proposals	2 – 25	=MIN(Proposals) and =MAX(Proposals)
Sales	1 – 15	=MIN(Sales) and =MAX(Sales)
FYP	₹8,000 – ₹60,000	=MIN(FYP) and =MAX(FYP)

5.2 Handling Missing Values

Missing values (blank cells) are a common issue in real-world HR datasets. In this project, the following approach was used to handle them:

- Activity Log: Since values are formula-generated (RANDBETWEEN), there are no missing values in the core columns. However, some auxiliary columns had blank cells — these were left blank intentionally or filled with 'N/A'.
- Agent Master: All 30 agent records were complete with no missing fields.

- Performance Sheet: Total FYP was calculated using SUMIF to aggregate across all months — if an agent had no records for a month, their total would still be correct (missing months simply contribute zero).
- Retention Sheet: The formula-based Risk column handles edge cases using nested IF statements, so missing/zero values are treated as high risk by design.

5.3 Creating Calculated Columns

Several new columns were added to the dataset to enable deeper analysis. These are the most important additions:

Conversion Rate Columns (Activity Log)

Three conversion rate columns were added to measure efficiency at each stage of the sales funnel:

Column Name	Excel Formula	What It Measures
Meeting Rate	=Meetings / Calls	What % of calls result in a meeting (target: >20%)
Proposal Rate	=Proposals / Meetings	What % of meetings result in a proposal (target: >30%)
Sales Conversion	=Sales / Proposals	What % of proposals result in a sale (target: >30%)

Problem Identification Column (Activity Log)

A nested IF formula was used to automatically label each record with the biggest problem area:

=IF(Meeting Rate < 0.2, "Poor Communication", IF(Proposal Rate < 0.3, "Poor Closing", IF(Sales Conversion < 0.3, "Poor Pitching", "Strong Performer")))

This formula works in order of priority — it first checks communication, then closing, then pitching. If none of the conditions is met, the agent is flagged as a Strong Performer.

Performance Classification (Performance Sheet)

The PERCENTILE function was used to set thresholds for performance classification:

- =PERCENTILE(FYP column, 0.5) → Gives the 50th percentile (median) as the lower threshold
- =PERCENTILE(FYP column, 0.9) → Gives the 90th percentile as the upper threshold for Star performers

These thresholds were then used in an IF formula: =IF(Total FYP >= 90th percentile, 'star', IF(Total FYP >= 50th percentile, 'average', 'low'))

Retention Risk Column (Retention Sheet)

A nested IF formula was used to assign risk levels based on last month's activity: =IF(AND(Meetings < 5, Sales = 0), 'High Risk', IF(Meetings < 10, 'Medium Risk', 'Low Risk'))

6. Productivity Analysis

Productivity analysis in insurance HR is all about understanding how efficiently agents convert their effort (calls) into revenue (sales and FYP). We do this by analysing the activity funnel — the step-by-step process from initial contact to final sale.

6.1 The Activity Funnel

The activity funnel is a simple but powerful concept. Think of it as a water funnel — you pour a lot of activity in at the top, but only a fraction comes out as sales at the bottom. Each step of the funnel is a conversion opportunity.

The Insurance Agent Activity Funnel

- STEP 1 — CALLS: Agent reaches out to potential customers (70 to 160 calls per month)
- STEP 2 — MEETINGS: Some calls result in a meeting (5 to 50 meetings per month)
- STEP 3 — PROPOSALS: Some meetings result in a proposal being sent (2 to 25 proposals)
- STEP 4 — SALES: Some proposals result in a policy being sold (1 to 15 sales per month)
- STEP 5 — FYP: Sales generate revenue in the form of First Year Premium (₹8,000–₹60,000)

The funnel naturally narrows at each stage, which is normal. The goal is to minimise the drop-off at each stage by improving agent skills and processes.

6.2 Conversion Rate Calculations

Based on the ranges in our dataset, we can calculate approximate average conversion rates for a typical agent:

Funnel Stage	Average Input	Average Output	Conversion Rate	Target Rate
Calls → Meetings	115 calls	27 meetings	~23.5%	> 20%
Meetings → Proposals	27 meetings	13 proposals	~48.1%	> 30%
Proposals → Sales	13 proposals	8 sales	~61.5%	> 30%
Sales → FYP	8 sales	₹34,000 FYP	—	Maximise

Looking at these numbers, the rates seem acceptable on average. However, averages can be misleading. The problem classification data tells us that many agents fall significantly below these averages — particularly at the communication stage (calls to meetings), where 46 out of 130 records were flagged as 'Poor Communication'.

6.3 Funnel Leakage Analysis

Funnel leakage refers to the point in the funnel where the biggest drop-off occurs. By examining our Insight → Action sheet data, we can quantify leakage:

Problem Type	Count	% of Total	Funnel Stage Affected
Poor Communication	46	35.4%	Calls → Meetings (most critical leakage)
Poor Closing	31	23.8%	Meetings → Proposals (second biggest leak)
Poor Pitching	17	13.1%	Proposals → Sales (smaller but important leak)
Strong Performer	36	27.7%	No significant leakage

The data clearly shows that the biggest productivity leak is at the very first conversion stage — getting meetings from calls. This is a communication and interpersonal skills problem. Almost 35% of all agent-month records show this weakness, making it the top training priority.

The second biggest leak is at the closing stage. These agents are doing a good job of getting meetings but are struggling to convert the interest into formal proposals. This suggests a need for training in consultative selling and needs assessment.

The third stage (pitching) has a lower leak rate, but it is still important because by the time an agent reaches the proposal stage, a lot of effort has already been invested. Losing a deal at this stage is particularly costly.

7. Agent Performance Segmentation

Performance segmentation is the process of dividing agents into groups based on their results. This allows managers to apply different strategies to each group rather than treating all agents the same way.

7.1 The Percentile Method

In this project, we use the percentile method to segment agents. A percentile is a statistical measure that tells you where a specific value stands relative to all other values in the dataset.

For example, if an agent is at the 90th percentile, it means they performed better than 90% of all agents. If they are at the 50th percentile, they are exactly in the middle of the group.

We used two percentile thresholds calculated using Excel's PERCENTILE function:

- 50th Percentile (Median) = ₹1,52,424 → Agents below this are 'Low Performers'
- 90th Percentile = ₹1,93,090 → Agents at or above this are 'Star Performers'
- Between 50th and 90th = 'Average Performers'

The advantage of using percentiles over fixed thresholds is that they automatically adjust to the dataset. If the overall performance level of all agents improves, the thresholds move up accordingly, always keeping the top 10% as 'Stars'.

7.2 Performance Classification Results

After applying the percentile formula to all 30 agents' Total FYP data, the following classification was obtained:

Performance Category	Number of Agents	% of Total Agents	FYP Range
Star Performer	3	10%	₹1,93,090 and above
Average Performer	12	40%	₹1,52,424 to ₹1,93,089
Low Performer	15	50%	Below ₹1,52,424

Star performers in our dataset include agents like A107 (₹2,14,870 FYP), A105 (₹1,95,648 FYP), and A116 (₹1,94,258 FYP). These three agents are the champions of the team. Low performers include agents like A126 (₹87,974 FYP) and A118 (₹89,791 FYP), whose FYP is less than half of the top performers.

7.3 The Pareto Principle (80/20 Rule) in Insurance

The Pareto Principle, also known as the 80/20 rule, states that in many situations, 80% of the results come from 20% of the effort or contributors. In the insurance industry, this often translates to: 80% of the revenue is generated by 20% of the agents.

In our dataset, while we have only 10% star performers, the Pareto concept still applies directionally. The top 10% of agents (3 agents) generate significantly above-average FYP, while the bottom 50% (15 agents) generate well below the median.

The practical implication of the Pareto principle for insurance HR management is:

- Protect your top performers at all costs — losing even one star agent has an outsized impact on revenue
- Focus development resources on the 40% of average performers who have the potential to move up
- Identify whether low performers have a solvable problem (skill gap) or an unsolvable one (wrong fit)

Key Insight from Performance Analysis

- 50% of agents are Low Performers — showing a significant productivity gap in the team
- Only 10% of agents are Star Performers — a very small group driving business outcomes
- 40% of agents are Average Performers — the biggest opportunity for growth through coaching
- Agent A107 is the top performer with ₹2,14,870 in FYP across the analysis period

8. Retention Analysis

Retention analysis is the process of identifying which agents are at risk of leaving the company and taking proactive steps to prevent their departure. Retaining good agents is critical because recruiting and training a new agent takes time and money.

8.1 Why Retention Matters

According to industry estimates, replacing an insurance agent can cost anywhere from 50% to 200% of their annual earnings — including recruitment costs, training costs, and the lost revenue during the transition period. Therefore, retaining agents, especially medium-risk ones, is a high-priority HR task.

The warning signs of an agent who is about to leave are often visible in their activity data before they formally resign. A drop in meetings scheduled, a fall in sales, reduced responsiveness — these are all leading indicators of disengagement.

8.2 Risk Classification Logic

In this project, we classify each agent's retention risk based on two variables from the most recent month:

- Meetings Last Month: How many meetings did the agent schedule?
- Sales Last Month: How many policies did the agent sell?

The classification logic (implemented as an Excel nested IF formula) works as follows:

Risk Level	Condition	What It Means	Action Required
High Risk	Meetings < 5 AND Sales = 0	Agent is nearly inactive — very likely to leave	Urgent call from manager within 24 hours
Medium Risk	Meetings < 10 (and not High Risk)	Agent is below average activity — at risk	Weekly check-in, training offer, incentive
Low Risk	Meetings ≥ 10	Agent is active and engaged — likely to stay	Regular recognition and motivation

8.3 Retention Analysis Results

The Retention sheet contains 130 agent-month records. The risk distribution is as follows:

Risk Category	Count of Records	Percentage	Insight
Low Risk	114	87.7%	Majority of agents are active and engaged
Medium Risk	16	12.3%	Requires close monitoring and support
High Risk	0	0%	No critically at-risk agents currently

The good news is that there are no High Risk agents in our current dataset. However, the 16 Medium Risk records (12.3%) should not be ignored. Some agents like A130 (with only 3 meetings in one month) and A109 (with only 4 meetings) are close to entering the high-risk zone.

8.4 The Retention Risk Cycle

It is important to understand that retention risk is not static — it can change every month. An agent who is Low Risk today can become Medium Risk if they have a bad month, and Medium Risk can quickly escalate to High Risk if no action is taken. The retention analysis should therefore be run every month to get the current picture.

The recommended monitoring cycle is:

- Monthly: Run the retention analysis for all agents using the previous month's data
- Weekly: For all Medium Risk agents, have the manager conduct a brief check-in call
- Daily: For any High Risk agent (if they appear), the branch manager should reach out personally

9. Training and Development Insights

Training is not just about teaching agents new skills — it is about fixing specific gaps identified through data. HR analytics makes training more targeted and therefore more effective.

9.1 The Importance of Training in Insurance

Insurance agents, unlike many other professionals, need to excel at multiple skills simultaneously:

- **Communication Skills:** To engage potential customers during cold calls and build rapport
- **Product Knowledge:** To accurately explain policy benefits, premiums, and terms
- **Consultative Selling:** To understand the customer's needs and match them with the right product
- **Objection Handling:** To address customer concerns and overcome their hesitations
- **Closing Techniques:** To move the conversation from interest to commitment

Weakness in any one of these areas can create a bottleneck in the sales funnel. This is exactly what our activity log data measures — and it tells us precisely where each agent needs help.

9.2 Data-Driven Training Recommendations

Based on the problem identification analysis, the following targeted training programs are recommended:

Problem Identified	Training Program	Target Agents	Expected Outcome
Poor Communication (46 cases)	Cold Call Mastery & Rapport Building Workshop	Agents with Meeting Rate < 20%	Improve calls-to-meetings conversion by 15-20%
Poor Closing (31 cases)	Consultative Selling & Needs Assessment Training	Agents with Proposal Rate < 30%	Improve meetings-to-proposal conversion by 10-15%
Poor Pitching (17 cases)	Presentation & Closing Techniques Workshop	Agents with Sales Conversion < 30%	Improve proposal-to-sales conversion by 10%
Low Overall FYP	High-Ticket Policy Sales Training	All Low Performers	Increase average FYP per sale

9.3 The Mentor-Buddy System

In addition to formal training, a mentor-buddy system can be highly effective. In this system:

- Each star performer is paired with two or three low or average performers
- The star performer shares their approach to calls, meetings, and closings
- Weekly meetings between mentor and mentee track progress and address challenges

This approach leverages the institutional knowledge of top performers and creates a culture of continuous learning within the team.

9.4 Measuring Training Effectiveness

For training to be useful, its impact must be measurable. The following KPIs should be tracked before and after training:

- Meeting Rate: Did more calls result in meetings after the communication training?
- Proposal Rate: Did more meetings result in proposals after the consultative selling training?
- Sales Conversion: Did more proposals convert to sales after the closing training?
- Total FYP: Did the agent's overall revenue generation improve?

By comparing these metrics before and after training, HR can calculate the Return on Investment (ROI) of each training programme.

10. Dashboard Design

A dashboard is a visual summary of the most important information in the project. It is designed for busy managers who need a quick overview without reading detailed reports. In this project, the dashboard was built entirely in Microsoft Excel.

10.1 Dashboard Design Philosophy

The dashboard was designed with three principles in mind:

- **Simplicity:** Only the most important information is shown — no clutter
- **Clarity:** Charts and numbers are clearly labelled so anyone can understand them
- **Actionability:** Every element on the dashboard should answer a 'So what?' question — what action does this information call for?

10.2 Key Performance Indicators (KPIs)

KPIs are the most important numbers that summarise the overall health of the agent workforce. The following KPIs were selected for the dashboard:

KPI Name	Excel Formula / Source	Value	Business Meaning
Total Agents	=COUNTA(AgentID column)	30	Total unique agents in the workforce
Total FYP Generated	=SUM(Total FYP column)	₹45,09,800	Total revenue from all agents
Average FYP per Agent	=AVERAGE(Total FYP)	₹1,50,327	Typical revenue per agent
Star Performers	Count of 'star' classification	3 (10%)	Top revenue generators
Medium Risk Agents	Count of Medium Risk records	16 (12.3%)	Agents needing attention
Poor Communication %	Count/Total	35.4%	Biggest training priority

10.3 Charts Used in the Dashboard

The dashboard uses four types of charts, each serving a specific analytical purpose:

Chart 1: Pie Chart — Performance Distribution

This chart shows the split between Star (10%), Average (40%), and Low (50%) performers. It immediately communicates the skewed nature of performance and the large proportion of low performers. The three colours used are Green (Star), Orange (Average), and Red (Low) — a traffic light colour scheme that is intuitive.

Chart 2: Bar Chart — Agent Problem Distribution

This chart shows how many agents fall into each problem category (Poor Communication, Poor Closing, Poor Pitching, Strong Performer). The bars are sorted in descending order so the biggest problem stands out immediately. This chart is the main input for training decisions.

Chart 3: Bar Chart — Retention Risk Distribution

This chart shows the breakdown of Low Risk vs. Medium Risk agents. The low proportion of medium-risk agents is reassuring, but the chart makes it easy to track this number month over month. If the Medium Risk bar starts growing, it is an early warning for the HR team.

Chart 4: Top Agents Performance Chart

This bar chart lists the top agents by Total FYP, making it easy to identify the star performers and their relative performance. It is also useful for recognition and reward decisions — the chart answers the question 'Who should receive a bonus or award this quarter?'

10.4 Dashboard Layout and Design

The dashboard was laid out in a clean single-page view in Excel. The layout follows a Z-pattern reading order (top-left to top-right, then bottom-left to bottom-right):

- Top Row: Four KPI cards showing total agents, total FYP, star performer count, and medium risk count
- Middle Left: Performance distribution pie chart
- Middle Right: Problem distribution bar chart
- Bottom: Top agent performance table and retention risk bar chart

11. Key Insights

Based on the complete analysis of all four datasets — Agent Master, Activity Log, Performance, and Retention — the following key insights have been identified:

11.1 Performance Distribution is Heavily Skewed

The most striking finding from the performance analysis is that exactly 50% of agents fall in the low performer category. This is not a minor issue — it means half the workforce is generating less than the median revenue. The company is essentially carrying a significant under-performing segment.

Meanwhile, only 3 agents (10%) are classified as star performers. These three individuals are producing significantly above-average FYP and are critical to the team's overall revenue. Their departure would have a disproportionate negative impact.

11.2 Communication is the Biggest Sales Bottleneck

The funnel leakage analysis clearly shows that Poor Communication is the single biggest problem, affecting 46 out of 130 agent-month records (35.4%). This means that the most common reason agents fail to generate more sales is not because of poor product knowledge or weak closing skills, but because they cannot convert enough calls into meetings.

This has an important implication for training priorities: before teaching agents how to close deals, we need to teach them how to open conversations.

11.3 The Majority of the Workforce is Stable

On a positive note, 87.7% of agent-month records are classified as Low Risk in the retention analysis. This suggests that most agents are currently engaged and active. The company is not in a crisis situation, but the 12.3% Medium Risk records require attention to prevent escalation.

11.4 No High-Risk Agents — But Vigilance is Needed

There are currently no High Risk agents in the dataset. However, several agents are close to the boundary (e.g., A130 with only 3 meetings in one month). Regular monthly monitoring is essential to catch these agents before they cross into the high-risk zone.

11.5 Agent Recruitment Source Insights

From the Agent Master data, the three recruitment sources are Online, Walk-in, and Referral. A full breakdown of active vs. inactive agents by source would reveal which recruitment channel produces the best quality of agent. This is important for optimising future hiring spend.

Summary of Key Insights

- 50% of agents are Low Performers — significant productivity improvement opportunity
- Only 10% are Star Performers — small but critical group driving revenue
- 35.4% of activity records show Poor Communication — top training priority
- 12.3% of records are Medium Risk — needs monitoring to prevent attrition escalation
- 0 High Risk agents currently — workforce is relatively stable
- 3 star agents (A107, A105, A116) should be prioritised for retention programmes

12. Recommendations

Based on the insights derived from the data analysis, the following actionable recommendations are made to the HR management team and branch managers:

12.1 Training for Low Performers

The 15 low-performing agents need immediate and structured support. A blanket training programme will not work — the training must be personalised based on each agent's specific problem area as identified in the Activity Log.

- Agents with Poor Communication: Enrol in a 2-day communication and cold-calling workshop. Focus on opening scripts, handling initial rejections, and building rapport in the first 60 seconds of a call.
- Agents with Poor Closing: Provide training in consultative selling — how to identify the customer's needs and match them with the right policy, rather than trying to sell a product.
- Agents with Poor Pitching: Give presentation skills training and role-play exercises. Practice handling common objections (price, trust, urgency).
- All Low Performers: Pair each low performer with a star performer as a buddy or mentor for at least 3 months. Weekly check-ins should be mandatory.

12.2 Retention Strategies for Top Performers

The three star performers (A107, A105, A116) are the company's most valuable HR assets. Losing even one of them would significantly impact revenue. The following retention strategies are recommended:

- Financial Incentives: Ensure star performers are receiving competitive commissions, bonuses, and performance-linked pay. Benchmark their earnings against the industry.
- Recognition: Publicly recognise star performers in team meetings, branch newsletters, and annual awards. People stay where they feel valued.
- Career Growth: Discuss long-term career paths — can a star performer become a team leader, trainer, or regional manager? Growth opportunities are a powerful retention tool.
- Flexible Working: Offer star performers flexibility in how they manage their time and territory. High performers often resent micromanagement.
- Retention Bonuses: Consider a retention bonus structure where star performers receive an additional reward for every year they stay with the company.

12.3 Monitoring Medium-Risk Agents

The 16 Medium Risk records represent agents who are showing signs of disengagement. These are the agents most likely to become High Risk in the next 1–2 months if nothing is done. The recommended approach is:

- Weekly Manager Check-in: Each medium-risk agent should receive a brief (15-minute) check-in call from their manager every week. The call should be supportive, not interrogative.
- Goal Setting: Help medium-risk agents set small, achievable weekly goals (e.g., 'Make 15 calls and aim for 3 meetings this week'). Small wins rebuild confidence.
- Root Cause Investigation: Find out why the agent's activity dropped. Was it a personal issue? A difficult territory? Lack of product knowledge? The solution depends on the cause.

- **Training Offer:** Proactively offer training to medium-risk agents. Frame it as an opportunity, not a penalty — 'We noticed you've had a tough few weeks. We want to help.'
- **Peer Support:** Connect medium-risk agents with high-performing peers who can share tips and encourage them.

12.4 Monthly HR Analytics Review

To make HR analytics a sustainable practice (not a one-time exercise), the company should establish a monthly review process:

- Run the retention analysis at the start of each month using the previous month's data
- Update the performance classification every quarter (as more FYP data accumulates)
- Review the problem distribution chart monthly to check if training programmes are working
- Share key metrics (KPIs from the dashboard) with all branch managers in a monthly HR meeting

13. Limitations of the Study

Like all research projects, this study has certain limitations that should be acknowledged. Understanding these limitations helps in interpreting the findings correctly and planning improvements.

13.1 Limited Dataset Size

The dataset contains only 30 unique agents and 130 activity records spanning approximately 5 months. This is a relatively small sample. In a real-world insurance company, there could be hundreds or thousands of agents across multiple branches and regions. Conclusions drawn from a small dataset may not generalise to larger populations.

For example, our finding that 10% are star performers is based on only 3 agents. With a larger dataset, the proportion might be different, and the percentile thresholds would be more reliable.

13.2 Simulated Data (RANDBETWEEN Formulas)

The activity data in the dataset is generated using Excel's RANDBETWEEN function rather than being based on actual historical records. This means:

- The data does not reflect real agent behaviour patterns — actual agents might show more consistent or more variable performance
- Seasonal trends (which are very common in insurance — e.g., higher sales at year-end for tax-saving products) are not captured
- The random nature of the data means the results will change every time the worksheet is refreshed

13.3 Basic Excel-Based Analysis

This project uses Microsoft Excel for all analysis, which is appropriate for a beginner-level project. However, Excel has several limitations compared to more advanced tools:

- No predictive modeling: We can classify current risk, but we cannot build a statistical model that predicts the probability of an agent leaving
- Limited visualisation options: Excel charts are functional but limited in interactivity and design compared to tools like Power BI or Tableau
- Scalability issues: Excel becomes slow and unwieldy with very large datasets (tens of thousands of rows)
- No automation: The analysis must be manually refreshed and reviewed each month — there is no automated alert system

13.4 Single Performance Metric

Our performance classification uses only Total FYP as the measure of performance. However, a truly comprehensive performance evaluation would consider multiple factors including:

- Customer retention rate (are the policies renewing in year 2 and beyond?)
- Claim ratio (are the agents selling to quality customers?)
- Number of new customers acquired vs. existing customers
- Compliance and documentation quality

13.5 No External Benchmarking

Our analysis is entirely internal — we compare agents within our own team but do not compare against industry benchmarks. It is possible that even our 'star performers' are below the industry average for top agents. Access to industry benchmark data would significantly enhance the value of the analysis.

14. Conclusion

This project successfully demonstrates how HR analytics can be applied to an insurance company's agent management problem using simple but powerful tools available in Microsoft Excel. Starting from raw agent data, the project built a complete analytical pipeline — from data cleaning to performance segmentation to retention risk classification — and extracted meaningful, actionable insights.

14.1 Summary of Findings

The key findings of this project can be summarised as follows:

- The 30-agent workforce shows a classic performance distribution: 50% low performers, 40% average performers, and only 10% star performers. This is consistent with the Pareto principle observed across many industries.
- Communication is the biggest skill gap: 35.4% of activity records show poor call-to-meeting conversion, making it the highest priority area for training investment.
- The retention risk profile is relatively healthy: 87.7% of agent records are Low Risk. However, 16 Medium Risk records (12.3%) require proactive manager attention.
- The activity funnel (Calls → Meetings → Proposals → Sales → FYP) is a powerful framework for understanding where individual agents lose productivity.
- The three star performers (A107, A105, A116) are critical assets that must be actively retained through financial incentives, recognition, and career growth opportunities.

14.2 Importance of HR Analytics in Decision-Making

Before HR analytics, managers in insurance companies would typically manage by intuition — promoting agents who seemed confident, training everyone on the same topics, and only noticing retention problems when an agent resigned. This reactive approach is expensive and ineffective.

HR analytics transforms this reactive approach into a proactive one:

- Instead of guessing who needs training, we know exactly which agents have which specific skill gaps
- Instead of waiting for agents to resign, we identify retention risk 1–2 months in advance
- Instead of treating all agents the same, we provide personalised support based on data

Even at this basic Excel level, the insights from this project are actionable and valuable. A branch manager who receives this analysis every month can make significantly better decisions about training allocation, manager attention, and retention initiatives.

This project also demonstrates that HR analytics does not require expensive software or advanced coding skills. With a well-structured Excel workbook and a clear analytical framework, even a small HR team or student can produce insights that drive real business value.

15. Future Scope

While this project has achieved its objectives as a beginner-level Excel-based analysis, there are many exciting opportunities to enhance and extend the work using more advanced tools and techniques.

15.1 Transition to Advanced Analytics Tools

Microsoft Power BI

Power BI is a business intelligence tool from Microsoft that connects directly to Excel data and creates interactive dashboards. With Power BI, the static charts in our Excel dashboard could become:

- Interactive filters: Click on a region and see only that region's performance data
- Drill-down capabilities: Click on 'Low Performers' to see the individual agents and their details
- Automatic refresh: Connect to a live database and the dashboard updates automatically
- Mobile access: Managers can view the dashboard on their phones

Python (Pandas, Matplotlib, Seaborn)

Python is a programming language widely used in data analytics. Using Python libraries, the same analysis done in Excel can be automated and scaled to handle thousands of agents:

- Pandas: For data cleaning, transformation, and aggregation (equivalent to Excel's formulas and pivot tables)
- Matplotlib / Seaborn: For creating professional statistical charts (histograms, box plots, heat maps)
- Scikit-learn: For machine learning models that can predict retention risk with much greater accuracy

15.2 Predictive Modeling Improvements

The retention risk model used in this project is rule-based — it classifies agents into risk categories based on fixed thresholds. A more advanced approach would use machine learning to predict the probability of attrition:

- Logistic Regression: A statistical model that predicts the probability (0% to 100%) that an agent will leave in the next 30 days
- Random Forest: A more complex model that can identify non-linear relationships between features (e.g., the combination of low meetings AND declining FYP trend is a stronger predictor than either alone)
- Feature Engineering: Adding new predictive variables such as time since last sale, number of consecutive months with declining activity, and manager interaction frequency

15.3 Expanded Dataset and Real-World Data

The current project uses simulated data. Future versions should use real historical data which would include:

- Multiple years of data to identify seasonality patterns (e.g., sales spikes around tax-saving season)
- Actual attrition records (who left, when, and why — through exit interview data)
- Product-level sales data (which policies are easiest/hardest to sell)
- Customer feedback data (satisfaction scores for policies sold by each agent)

15.4 Real-Time Monitoring and Alerts

The current Excel-based system requires manual monthly updates. A future system could implement:

- Automated data pipelines that pull activity data from the company's CRM system every day
- Automated email alerts to managers when an agent crosses into the Medium or High Risk zone
- A live dashboard that updates in real-time as agents log their activity

15.5 Integration with HR Systems

The ultimate vision for this project is to integrate the analytics with the company's existing HR systems:

- HRMS (Human Resource Management System): Automatically pull attendance, leave, and performance data
- CRM (Customer Relationship Management): Get real-time sales and activity data
- LMS (Learning Management System): Track which training modules each agent has completed and correlate with performance improvement

Such integration would create a comprehensive 'People Intelligence Platform' that gives HR leaders a 360-degree view of every agent in real time.

Project Summary

- Project Title: HR Analytics for Insurance — Agent Performance, Productivity, and Retention Analysis
- Tools Used: Microsoft Excel (RANDBETWEEN, IF, PERCENTILE, SUMIF, Pivot Tables, Charts)
- Dataset: 30 agents, 130 activity records across 5 months
- Key Outcome: Actionable insights on performance segmentation, productivity gaps, and retention risk
- Recommended Next Step: Implement monthly review process and enrol low performers in targeted training

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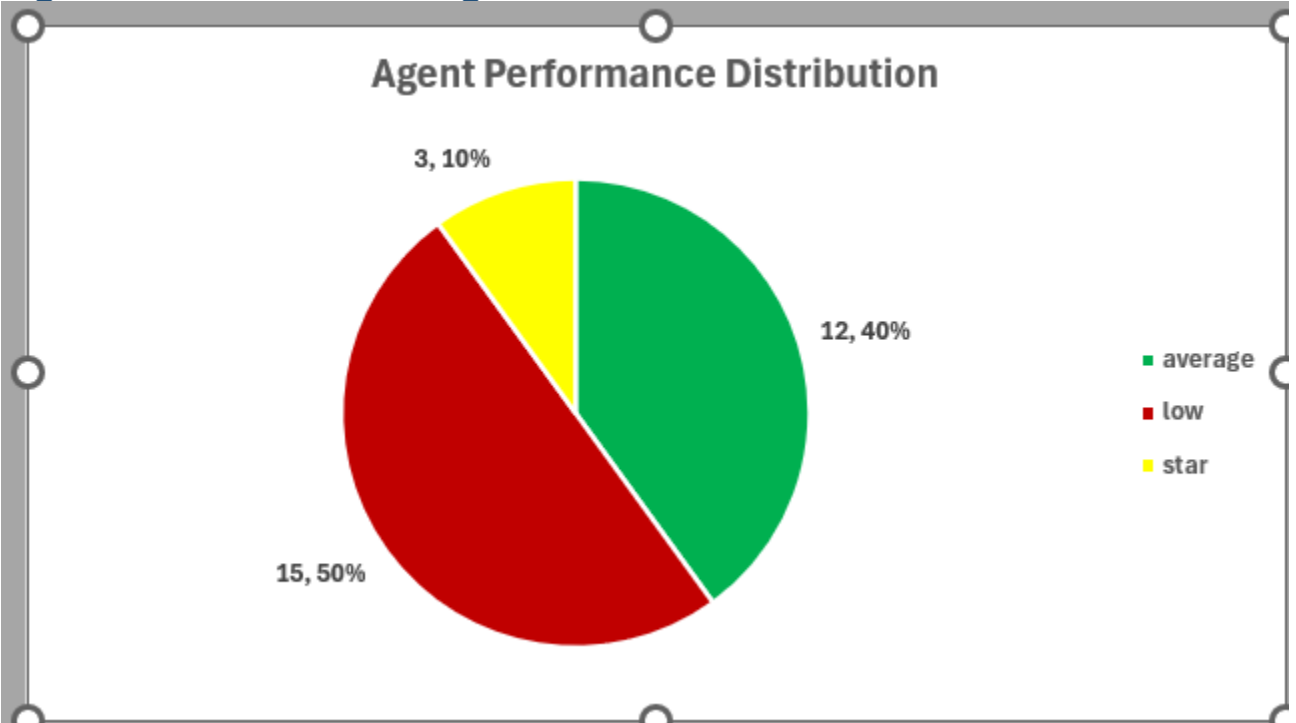
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Appendix: Dashboard and Analysis Screenshots

Dashboard Screenshot



Agent Performance Segmentation



Retention Analysis

AgentID	Meetings LastMonth	Sales LastMonth	Risk
A101	14	1	Low Risk
A102	20	2	Low Risk
A103	8	3	Medium Risk
A104	19	4	Low Risk
A105	9	5	Medium Risk
A106	34	6	Low Risk
A107	11	7	Low Risk
A108	22	8	Low Risk
A109	47	9	Low Risk
A110	40	10	Low Risk
A111	38	11	Low Risk
A112	20	12	Low Risk
A113	5	1	Medium Risk
A114	45	2	Low Risk
A115	22	3	Low Risk
A116	43	4	Low Risk
A117	13	5	Low Risk
A118	39	6	Low Risk
A119	4	7	Medium Risk
A120	22	8	Low Risk
A121	2	9	Medium Risk
A122	36	10	Low Risk
A123	10	11	Low Risk
A124	36	12	Low Risk

Row Labels

Count of AgentID

Low Risk114

Medium Risk16

Grand Total130

Low Risk88%

Medium Risk12%

Insights

"Agents with <5 meetings are at high risk"

✓ "Low activity is strong indicator of attrition"

Action

✓ Call high-risk agents (no risk in any agents)

✓ Give training

✓ Monitor weekly

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Activity Log

Insight → Action

Retention

Performance

Dashboard

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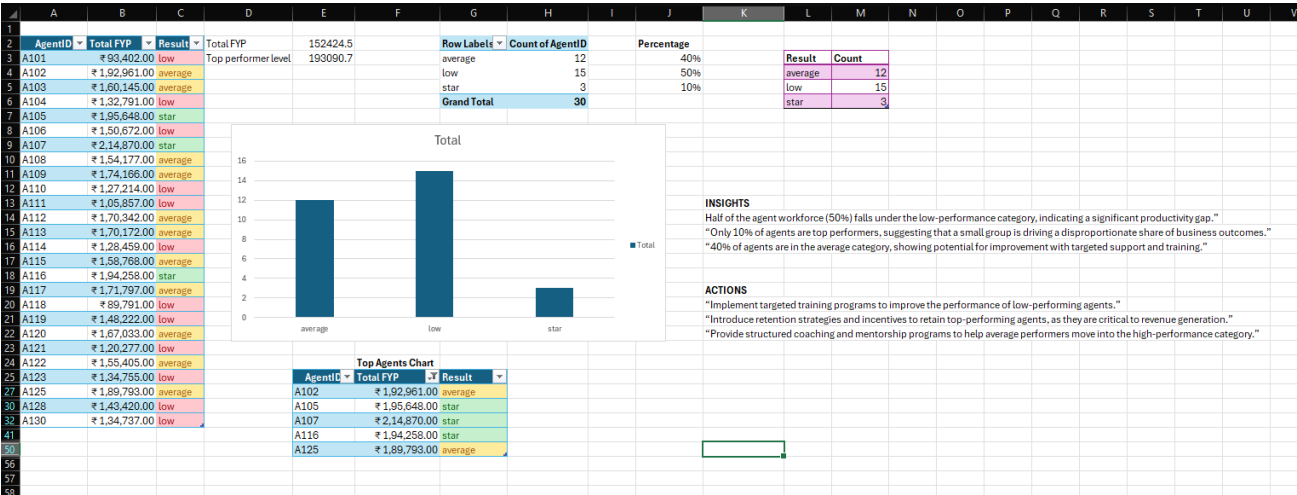
Retention Risk Analysis

Row Labels	Count of AgentID	Insights
Low Risk	114	"Agents with <5 meetings are at high risk"
Medium Risk	16	✓ "Low activity is strong indicator of attrition"
Grand Total	130	
Low Risk	88%	Action
Medium Risk	12%	✓ Call high-risk agents (no risk in any agents)
		✓ Give training
		✓ Monitor weekly

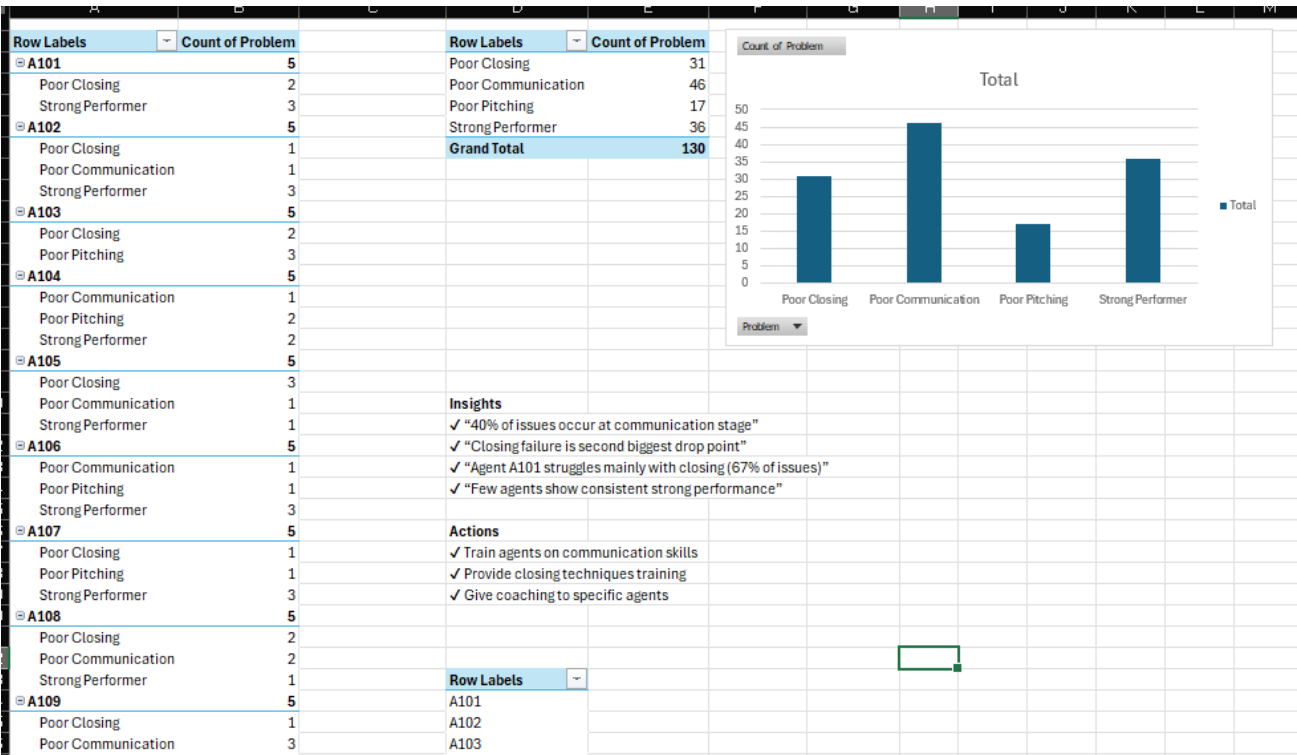
Agent Activity and Productivity Data



Performance



Insight → Action



— End of Report —